
MIDTERM EXAM 1 (FORM B)

NAME: SOLUTIONS

ID N°: _____

INSTRUCTIONS

- This is a closed-book exam. You may not use your notes, textbook or any outside resources to assist you.
- Write your answer to each question in the space assigned to each problem. If you want the instructor to read the scratch working page for the answer to a specific problem, explicitly write that on the space assigned to said problem.
- Always remember to show your work. An answer without a proper justification will earn little to no credit, even if it is correct.
- Circle or box your final answer to each question.
- Make sure that your answers are organized and legible. Use a dark pencil or a pen. Any work that is hard or impossible to read will not be graded.
- In this exam, the use of determinants is not allowed. We will allow determinants once we cover them in this course.
- The use of calculators of any type is not allowed.

This exam consists of four problems, and your total score will be out of $10 + 10 + 10 + 10 = 40$ points.

Good Luck!

This page is intentionally left blank

Problem 1 (10 points). Suppose A and B are 5×3 matrices.

- (a) (6 points) Find, in terms of A and B , the matrix X that satisfies

$$2(X - 3A^T)^T = 6X^T - B.$$

What is the size of X ?

$$2(X - 3A^T)^T = 6X^T - B \Rightarrow 2(X^T - 3A) = 6X^T - B \Rightarrow 2X^T - 6A = 6X^T - B \Rightarrow 4X^T = B - 6A$$

$$\Rightarrow X^T = \frac{1}{4}(B - 6A)$$

$$\Rightarrow \boxed{X = \frac{1}{4}(B^T - 6A^T)}$$

X is a 3×5 matrix because A^T and B^T are 3×5 matrices and X is the result of adding scalar multiples of A^T and B^T .

- (b) (4 points) Calculate the matrix $C^2 + C - 17I_2$, where

$$C = \begin{bmatrix} -2 & 3 \\ 5 & 1 \end{bmatrix}.$$

$$\begin{aligned} C^2 + C - 17I_2 &= \begin{bmatrix} -2 & 3 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} -2 & 3 \\ 5 & 1 \end{bmatrix} + \begin{bmatrix} -2 & 3 \\ 5 & 1 \end{bmatrix} - 17 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 18 & -3 \\ -5 & 16 \end{bmatrix} + \begin{bmatrix} -2 & 3 \\ 5 & 1 \end{bmatrix} + \begin{bmatrix} -17 & 0 \\ 0 & -17 \end{bmatrix} \\ &= \boxed{\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}} \end{aligned}$$

Problem 2 (2 points per part). On each of the following questions, **completely fill in** the circle next to the correct answer. Each question only has one correct answer. You do not need to justify your choices.

(1) If A is an $m \times n$ matrix, the rank r of A always satisfies:

- ☐ $r \leq m$ (but not necessarily $r \leq n$).
- ☐ $r \leq n$ (but not necessarily $r \leq m$).
- ☒ $r \leq n$ and $r \leq m$.
- ☐ None of the above.

(2) A system of linear equations is called *consistent* if:

- ☐ It has exactly one solution.
- ☐ It has infinitely many solutions.
- ☐ It has no solutions.
- ☒ It has at least one solution.

(3) If $A\mathbf{x} = \mathbf{b}$ is a *non-homogeneous* linear system and $\mathbf{x}_1, \mathbf{x}_2$ are solutions to it, then:

- ☐ $\mathbf{x}_1 + \mathbf{x}_2$ is a solution to $A\mathbf{x} = \mathbf{b}$.
- ☐ $\mathbf{x}_1 - \mathbf{x}_2$ is a solution to $A\mathbf{x} = \mathbf{b}$.
- ☐ $\mathbf{x}_1 + \mathbf{x}_2$ is a solution to $A\mathbf{x} = \mathbf{0}$.
- ☒ $\mathbf{x}_1 - \mathbf{x}_2$ is a solution to $A\mathbf{x} = \mathbf{0}$.

(4) Given the matrices $X = \begin{bmatrix} 1 & 0 & 2 \end{bmatrix}$ and $Y = \begin{bmatrix} 2 & 1 & 3 \end{bmatrix}^T$, we have:

- ☐ The matrices $5X + Y$, $X(2Y)$ and $4Y^T$ exist.
- ☒ The matrices XY , YX and $2X + 5Y^T$ exist.
- ☐ The matrices $Y(3X^T)$, $(X + Y^T)^T$ and Y^2 exist.
- ☐ The matrices $X^T X$, $-2X + Y$ and YX^2 exist.

(5) A homogeneous system of equations $A\mathbf{x} = \mathbf{0}$ has 20 variables and 8 equations. If the augmented matrix $[A \mid \mathbf{0}]$ has rank 5, then:

- ☐ The system is consistent regardless of A , and its general solution involves 12 parameters.
- ☒ The system is consistent regardless of A , and its general solution involves 15 parameters.
- ☐ There are matrices A for which the system is inconsistent. However, when the system is consistent, its general solution involves 12 parameters.
- ☐ There are matrices A for which the system is inconsistent. However, when the system is consistent, its general solution involves 15 parameters.

Problem 3 (10 points). Consider the system of linear equations $A\mathbf{x} = \mathbf{b}$ whose augmented matrix is:

$$[A|\mathbf{b}] = \left[\begin{array}{ccc|c} 1 & 0 & 6 & a \\ 3 & 1 & 5 & b \\ -4 & -2 & 2 & c \end{array} \right].$$

- (a) (2 points) What conditions do a , b and c have to satisfy in order for the system $A\mathbf{x} = \mathbf{b}$ to be homogeneous?

The system is homogeneous if and only if all constant terms are zero:

$$a=0, b=0, c=0$$

- (b) (4 points) Using the Gaussian algorithm, show that the system $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $c = 2a - 2b$.

$$\left[\begin{array}{ccc|c} 1 & 0 & 6 & a \\ 3 & 1 & 5 & b \\ -4 & -2 & 2 & c \end{array} \right] \xrightarrow[R_3+4R_1]{R_2-3R_1} \left[\begin{array}{ccc|c} 1 & 0 & 6 & a \\ 0 & 1 & -13 & b-3a \\ 0 & -2 & 26 & c+4a \end{array} \right] \xrightarrow{R_3+2R_2} \left[\begin{array}{ccc|c} 1 & 0 & 6 & a \\ 0 & 1 & -13 & b-3a \\ 0 & 0 & 0 & c+2b-2a \end{array} \right]$$

The system is (almost) in row-echelon form, so it is consistent if and only if it does not contain an equation of the form $0 = \text{nonzero}$. Thus:

$$\text{The system is consistent} \Leftrightarrow c+2b-2a=0 \Leftrightarrow c=2a-2b$$

(c) (4 points) Calculate a set of basic solutions to the system $A\mathbf{x} = \mathbf{0}$.

(**Hint:** Recycle the calculations from the previous part)

From the previous part (and setting $a=b=c=0$), we see that the system is equivalent to

$$\begin{cases} x + 6z = 0 \\ y - 13z = 0 \end{cases}$$

$\uparrow \quad \uparrow \quad \uparrow$
 Free variables $z=t$

The general solution is

$$\vec{x} = \begin{bmatrix} -6t \\ 13t \\ t \end{bmatrix} = t \begin{bmatrix} -6 \\ 13 \\ 1 \end{bmatrix}, \quad t \text{ arbitrary.}$$

Thus, a set of basic solutions is

$$\boxed{\vec{x}_1 = \begin{bmatrix} -6 \\ 13 \\ 1 \end{bmatrix}}$$

Problem 4 (10 points). Consider the matrix

$$A = \begin{bmatrix} 1 & -1 & 1 \\ -2 & 2 & -1 \\ 3 & -2 & 1 \end{bmatrix}.$$

(a) (6 points) Using elementary row operations, find the inverse A^{-1} of A .

$$\begin{aligned} & \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ -2 & 2 & -1 & 0 & 1 & 0 \\ 3 & -2 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow[R_3 - 3R_1]{R_2 + 2R_1} \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 & 1 & 0 \\ 0 & 1 & -2 & -3 & 0 & 1 \end{array} \right] \xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 1 & -2 & -3 & 0 & 1 \\ 0 & 0 & 1 & 2 & 1 & 0 \end{array} \right] \\ & \xrightarrow[R_1 - R_3]{R_2 + 2R_3} \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & -1 & -1 & 0 \\ 0 & 1 & 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 & 1 & 0 \end{array} \right] \xrightarrow{R_1 + R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 2 & 1 & 0 \end{array} \right] \\ & \text{So } A \text{ is invertible and } A^{-1} = \boxed{\begin{bmatrix} 0 & 1 & 1 \\ 1 & 2 & 1 \\ 2 & 1 & 0 \end{bmatrix}} \end{aligned}$$

(b) (4 points) Use the previous part to find the solutions to the system

$$\begin{cases} x - y + z = 2 \\ -2x + 2y - z = -1 \\ 3x - 2y + z = 3 \end{cases}$$

The matrix form of the system is $A\vec{x} = \vec{b}$, where $\vec{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $\vec{b} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$. As A is invertible, this system has exactly one solution:

$$\vec{x} = A^{-1}\vec{b} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 2 & 1 \\ 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} = \boxed{\begin{bmatrix} 2 \\ 3 \\ 3 \end{bmatrix}}$$

Scratch working page

Scratch working page